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Data Science Capstone - Ethics Component
Professor Hardin
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Faced and addressed

Disclosure risk

IPEDS data is aggregated, but at small colleges some rank/sex categories have so few people that the average starts to function almost like an individual salary. I addressed this by sticking to pre-aggregated, reasonably sized groupings and never trying to infer or back out an individual's pay from a narrow category.

Data integrity

The raw dataset mixes real subgroups with rollup "Total" rows and missing-data placeholder codes like total_months = -1. Left unfiltered, these would quietly distort any averages I computed. I addressed this through a deliberate cleaning pipeline, filtering out aggregate codes, deduping by staff count, and excluding placeholder values before making any comparisons.

Loaded framing

A phrase like "administrative bloat" carries a built-in value judgment, and a widening gap alone doesn't prove anyone is overpaid or unnecessary. I've tried to address this by keeping my language descriptive rather than accusatory, and by reporting my panel regression result honestly even though it came back statistically insignificant, rather than only surfacing analyses that fit a particular narrative.

Not yet faced

Presenting named Form 990 executives

This data names real individuals and their actual compensation, not institutional averages. It's legally public, but I haven't yet had to decide how to frame it, as evidence of a systemic pattern rather than a spotlight on specific people's pay.

Researcher positionality

My advisor is a college professor, meaning the side of my framing that looks "shortchanged" is the group my mentor belongs to. I haven't written my final conclusions yet, that's where this becomes relevant, so it's worth staying deliberately aware of once I move from reporting results to interpreting them.